

Srinath S

AI Engineer | Agentic AI | LangGraph | MCP | Python

+91 7603967977 | srinath2934@gmail.com | [LinkedIn](#) | [GitHub](#) | Bengaluru, India

Profile Summary

B.Tech Artificial Intelligence and Data Science graduate with strong CS fundamentals, DSA, and hands-on experience building agentic AI, RAG, NLP, and computer-vision systems. Built **ClosePilot**, a stateful 7-node LangGraph workflow with typed state, checkpoint-based Human-in-the-Loop (HITL) interrupts, safe pause/resume, Claude integration through MCP, structured-output validation, and a 14/14 passing Pytest suite. Also built **RepoChat**, a LangChain and FAISS retrieval system with automated response evaluation and LangSmith tracing. Portfolio & live project demos: srinath2934.github.io.

Professional Experience

Proitbridge Machine Learning Intern

Sep 2025 – Dec 2025

- Engineered a Python-based CatBoost classification pipeline using pandas and NumPy to process 100+ features from a consumer dataset, predicting loan-default risk with **90% accuracy and 0.74 ROC-AUC**.
- Optimized feature-selection and hyperparameter-tuning scripts, containerizing preprocessing and inference modules for integration testing on **AWS EC2**.
- Implemented **SHAP-based interpretability modules** in Python to explain feature contributions and communicate applicant-level predictions to business stakeholders.

Selected Engineering Projects

ClosePilot — Enterprise Multi-Agent AI Sales Copilot

[github](#) | [live demo](#)

- Architected a stateful **7-node LangGraph multi-agent system** (Research, Prioritize, Strategy, Communication, Approval, Action, Verification) with typed Pydantic v2 schemas and checkpoint memory for thread-level state persistence.
- Standardized CRM tool execution via **Model Context Protocol (MCP)** for HubSpot; enforced hard **Human-in-the-Loop (HITL) interrupt gates** (`interrupt_before=[ 'approval' ]`) to eliminate unauthorized autonomous CRM writes.
- Built deterministic deal priority ranking (`Stage + Value + Inactivity - Penalties`) with zero LLM variance; integrated Claude/NVIDIA LLM reasoning, LangSmith waterfall tracing, and verified reliability with a **14/14 passing Pytest suite**.

Semantic AI Support Ticket Intelligence Platform

[github](#) | **IEEE ICIRCA 2026**

- Engineered an operational triage backend processing **108,819 enterprise support tickets** across 4 multi-task outputs: category classification, routing queue, priority level, and resolution-time (ETA) regression.
- Implemented 384-dimensional **Sentence-BERT (all-MiniLM-L6-v2)** embeddings and **FAISS vector indexing** for sub-150ms semantic search over historical resolutions, preventing keyword-miss routing bottlenecks.
- Achieved **81% priority accuracy, 80% action accuracy, and 7.73-hour MAE on resolution ETA**; served via FastAPI with SQLite audit logging and co-authored a peer-reviewed paper accepted at **IEEE ICIRCA 2026**.

RepoChat — Semantic Repository Intelligence System (RAG)

[github](#)

- Built an enterprise RAG assistant that semantically parses, chunks, and indexes **1,000+ files per repository** (benchmarked on React and LangChain) into ChromaDB with **<300ms vector retrieval latency**.
- Engineered AST-aware semantic code chunking to preserve class and function scopes; integrated **Groq LPU (Llama 3.3 70B)** with exact line-number citations, achieving **~95% code-grounded accuracy and <2% hallucination rate**.
- Built a responsive Streamlit interface with session-state memory, vector store caching, visual repository browsing, and automated response evaluation.

Technical Skills

Machine Learning & AML – Supervised/Unsupervised Learning, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Classification, Regression, CatBoost, scikit-learn, SHAP, Hyperparameter Tuning

Deep Learning & NLP – PyTorch, Natural Language Processing (NLP), Sentence-BERT (SBERT), Vectorization, Tokenization & Chunking, Text Preprocessing, YOLOv9/v8, OpenCV, Multi-Head Architectures

LLMs & Retrieval (RAG) – Large Language Models (LLMs), Claude, OpenAI, Groq LPU, RAG Pipelines, FAISS Vector Indexing, ChromaDB, BAAI/BGE Embeddings, Context Compaction, Token Budgeting, Citations

Agentic Orchestration – LangGraph (v0.2+), State Machines, Typed State, Checkpointers, Human-in-the-Loop (HITL), MCP (Model Context Protocol), Tool Calling, Multi-Agent Workflows, CrewAI (explored)

Data Science & Systems – Python 3.10+ (async, typing, OOP), SQL (PostgreSQL, SQLite, MySQL), pandas, NumPy, Data Structures & Algorithms, FastAPI, Docker, AWS EC2, LangSmith Tracing, Pytest

Education

B.Tech – Artificial Intelligence & Data Science (GPA: 8.0 / 10)

2022–2026

JCT College of Engineering and Technology, affiliated with Anna University

HSC – Mathur Boys Higher Secondary School, Tamil Nadu State Board

2022

Certifications

- Oracle Certified Foundations Associate:** OCI 2023 AI Foundations (ID: 100662362OCI23AIFCA)
- Foundation: Introduction to LangChain – Python — LangChain Academy (Jul 2026)
- Introduction to Model Context Protocol — Anthropic
- Python (Intermediate) & SQL (Basic) Certification — HackerRank
- Python for Data Science, AI & Development — IBM via Coursera
- Databases and SQL for Data Science with Python — IBM via Coursera